

Access Control Policy Requirements

The following inter-department ACL policies must be implemented and verified. All policies are to be applied as extended named ACLs. Students must document the placement (interface, direction) of each ACL.

Source	Destination	Action	Policy rationale
VLAN_DEIE	VLAN_DIS	PERMIT all	Engineering staff have full server farm access
VLAN_DCEE	VLAN_DIS	PERMIT HTTP/HTTPS only	Admin staff may access web services in DIS only
VLAN_DMME	VLAN_DIS	DENY all	Workshop lab has no server farm access
VLAN_DCEE	VLAN_DEIE	DENY all	Admin must not access engineering subnet
VLAN_DMME	VLAN_DCEE	DENY all	Workshop is fully isolated from admin
VM_ZABBIX IP	All devices	PERMIT SNMP (UDP 161)	Monitoring agent must reach all devices
MGMT VLAN only	All devices	PERMIT SSH (TCP 22)	SSH management restricted to VLAN 99
Any	Any	DENY all (implicit)	Default deny - must be explicitly acknowledged

DEIE users inside the same vlan

```
DEIE-PC-01> ping 10.10.10.11

84 bytes from 10.10.10.11 icmp_seq=1 ttl=64 time=7.185 ms
84 bytes from 10.10.10.11 icmp_seq=2 ttl=64 time=4.854 ms
84 bytes from 10.10.10.11 icmp_seq=3 ttl=64 time=1.752 ms
84 bytes from 10.10.10.11 icmp_seq=4 ttl=64 time=6.173 ms
84 bytes from 10.10.10.11 icmp_seq=5 ttl=64 time=3.743 ms

DEIE-PC-01> trace 10.10.10.11
trace to 10.10.10.11, 8 hops max, press Ctrl+C to stop
 1  *10.10.10.11  2.450 ms (ICMP type:3, code:3, Destination port unreachable)

DEIE-PC-01> trace 10.10.10.10
traceroute to 10.10.10.10, 8 hops max
 1 10.10.10.10    0.001 ms
```

DEIE to DCEE

```
DEIE-PC-01> ping 10.10.20.10

*10.10.10.1 icmp_seq=1 ttl=255 time=9.177 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.10.10.1 icmp_seq=2 ttl=255 time=6.093 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.10.10.1 icmp_seq=3 ttl=255 time=7.791 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.10.10.1 icmp_seq=4 ttl=255 time=10.342 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.10.10.1 icmp_seq=5 ttl=255 time=6.655 ms (ICMP type:3, code:13, Communication administratively prohibited)

DEIE-PC-01> trace 10.10.20.10
trace to 10.10.20.10, 8 hops max, press Ctrl+C to stop
 1  *10.10.10.1  12.859 ms (ICMP type:3, code:13, Communication administratively prohibited) *
```

DEIE to DMME

```
DEIE-PC-01> ping 10.10.30.10

*10.10.10.1 icmp_seq=1 ttl=255 time=16.515 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.10.10.1 icmp_seq=2 ttl=255 time=7.164 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.10.10.1 icmp_seq=3 ttl=255 time=9.553 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.10.10.1 icmp_seq=4 ttl=255 time=6.892 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.10.10.1 icmp_seq=5 ttl=255 time=7.902 ms (ICMP type:3, code:13, Communication administratively prohibited)

DEIE-PC-01> trace 10.10.30.10
trace to 10.10.30.10, 8 hops max, press Ctrl+C to stop
 1  *10.10.10.1  10.525 ms (ICMP type:3, code:13, Communication administratively prohibited) *
```

DEIE to DIS

```
DEIE-PC-01> ping 10.10.40.10

84 bytes from 10.10.40.10 icmp_seq=1 ttl=62 time=46.652 ms
84 bytes from 10.10.40.10 icmp_seq=2 ttl=62 time=15.284 ms
84 bytes from 10.10.40.10 icmp_seq=3 ttl=62 time=13.376 ms
84 bytes from 10.10.40.10 icmp_seq=4 ttl=62 time=17.720 ms
84 bytes from 10.10.40.10 icmp_seq=5 ttl=62 time=11.356 ms

DEIE-PC-01> trace 10.10.40.10
trace to 10.10.40.10, 8 hops max, press Ctrl+C to stop
 1  10.10.10.1  6.957 ms  9.158 ms  7.818 ms
 2  10.255.0.9  13.752 ms  18.596 ms  15.275 ms
 3  *10.10.40.10  22.332 ms (ICMP type:3, code:3, Destination port unreachable)

DEIE-PC-01> trace 10.10.40.11
trace to 10.10.40.11, 8 hops max, press Ctrl+C to stop
 1  10.10.10.1  4.898 ms  7.565 ms  7.691 ms
 2  10.255.0.9  14.636 ms  16.564 ms  12.689 ms
 3  **10.10.40.11  11.663 ms (ICMP type:3, code:3, Destination port unreachable)
```

DCEE users inside the same vlan

```
DCEE-PC-01> ping 10.10.20.11

84 bytes from 10.10.20.11 icmp_seq=1 ttl=64 time=2.517 ms
84 bytes from 10.10.20.11 icmp_seq=2 ttl=64 time=3.511 ms
84 bytes from 10.10.20.11 icmp_seq=3 ttl=64 time=2.384 ms
84 bytes from 10.10.20.11 icmp_seq=4 ttl=64 time=3.281 ms
84 bytes from 10.10.20.11 icmp_seq=5 ttl=64 time=4.694 ms

DCEE-PC-01> trace 10.10.20.11
trace to 10.10.20.11, 8 hops max, press Ctrl+C to stop
 1  *10.10.20.11  6.116 ms (ICMP type:3, code:3, Destination port unreachable)

DCEE-PC-01> trace 10.10.20.10
traceroute to 10.10.20.10, 8 hops max
 1 10.10.20.10    0.001 ms
```

DCEE to DEIE

```
DCEE-PC-01> ping 10.10.10.10

*10.10.20.1 icmp_seq=1 ttl=255 time=6.902 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.10.20.1 icmp_seq=2 ttl=255 time=8.698 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.10.20.1 icmp_seq=3 ttl=255 time=4.596 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.10.20.1 icmp_seq=4 ttl=255 time=6.727 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.10.20.1 icmp_seq=5 ttl=255 time=18.954 ms (ICMP type:3, code:13, Communication administratively prohibited)

DCEE-PC-01> trace 10.10.10.10
trace to 10.10.10.10, 8 hops max, press Ctrl+C to stop
 1  *10.10.20.1  10.322 ms (ICMP type:3, code:13, Communication administratively prohibited) *
```

DCEE to DMME

```
DCEE-PC-01> ping 10.10.30.10

*10.10.20.1 icmp_seq=1 ttl=255 time=8.668 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.10.20.1 icmp_seq=2 ttl=255 time=4.694 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.10.20.1 icmp_seq=3 ttl=255 time=9.979 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.10.20.1 icmp_seq=4 ttl=255 time=7.411 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.10.20.1 icmp_seq=5 ttl=255 time=7.549 ms (ICMP type:3, code:13, Communication administratively prohibited)

DCEE-PC-01> trace 10.10.30.10
trace to 10.10.30.10, 8 hops max, press Ctrl+C to stop
 1  *10.10.20.1  10.038 ms (ICMP type:3, code:13, Communication administratively prohibited) *
```

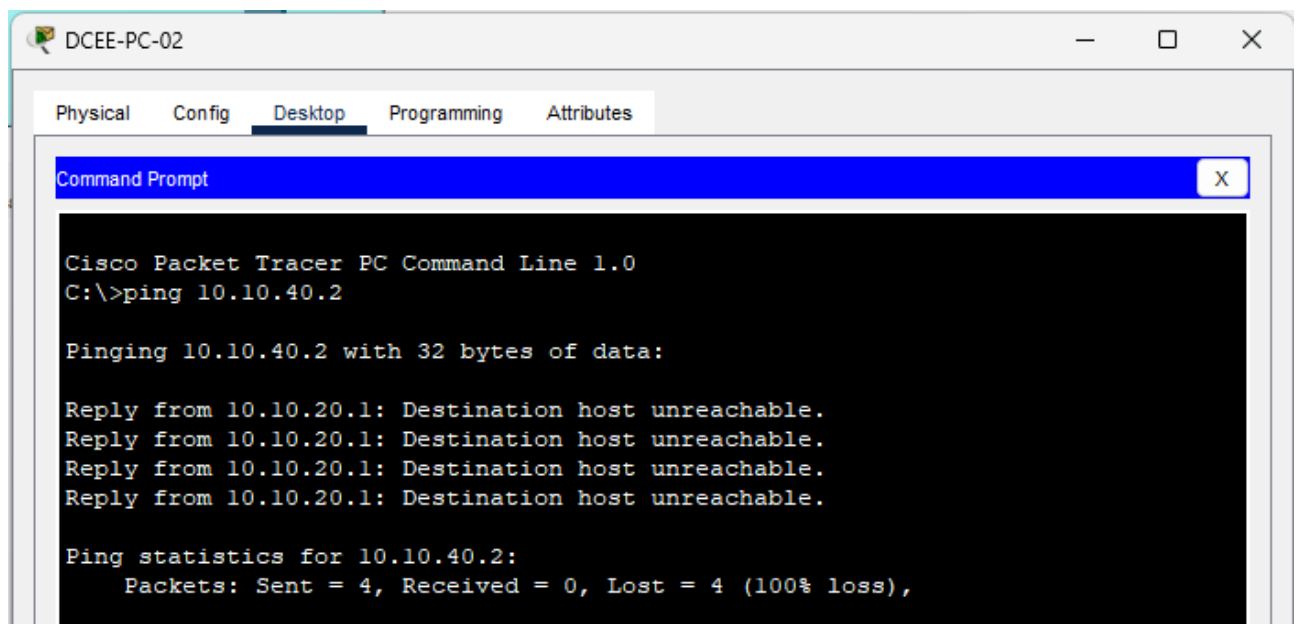
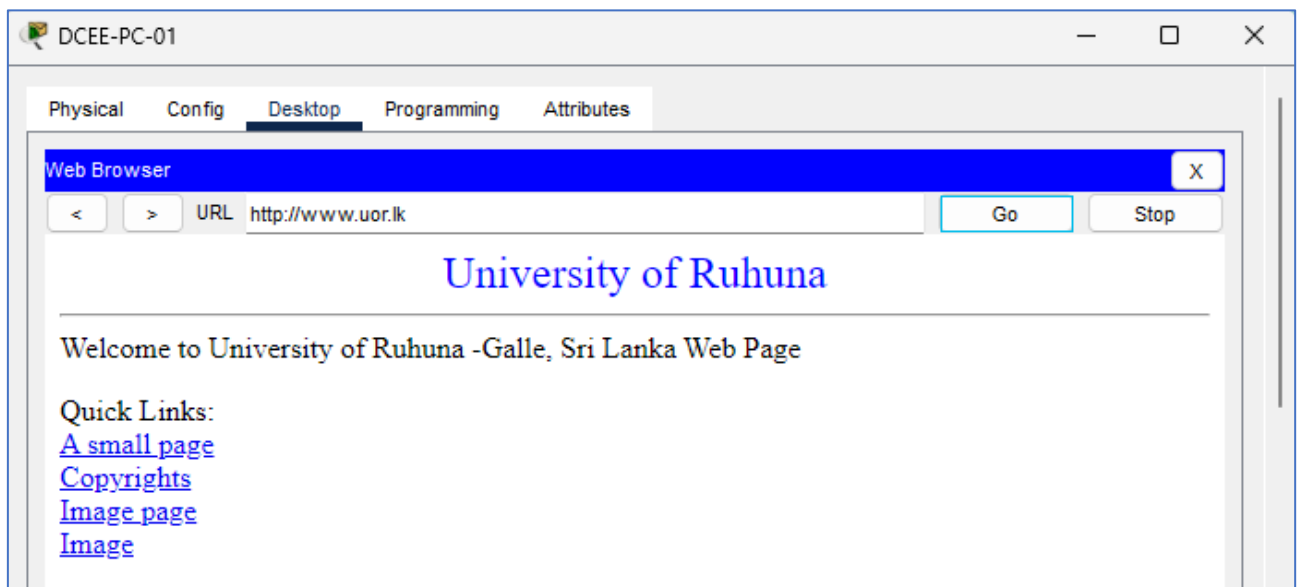
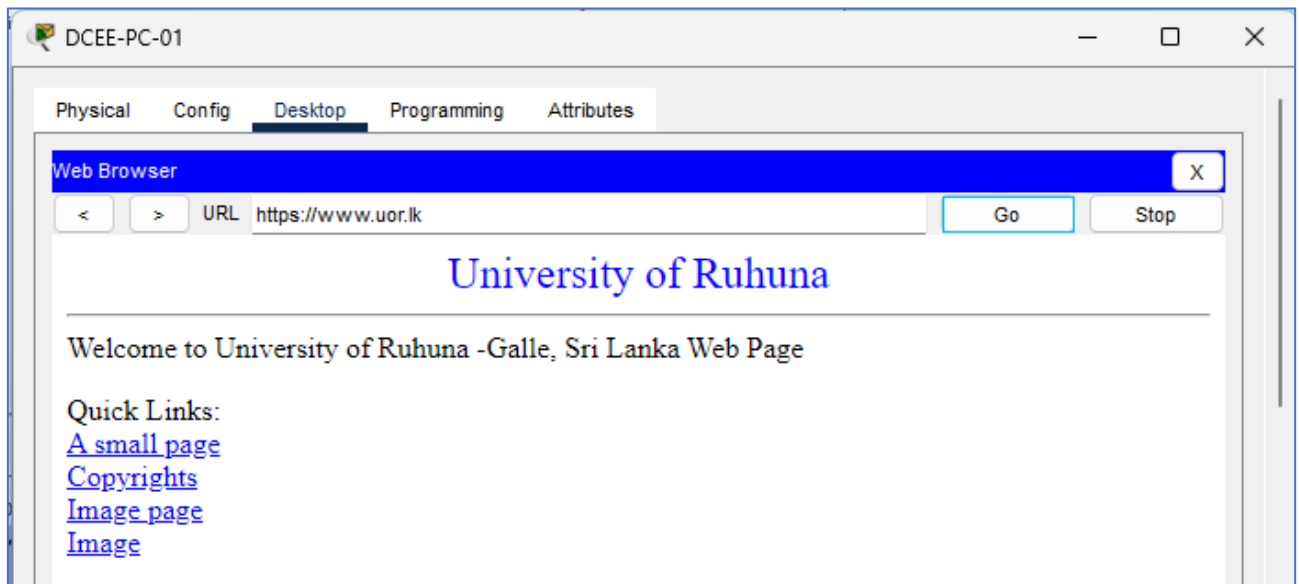
DCEE to DIS

```
DCEE-PC-01> ping 10.10.40.10

*10.10.20.1 icmp_seq=1 ttl=255 time=20.727 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.10.20.1 icmp_seq=2 ttl=255 time=11.870 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.10.20.1 icmp_seq=3 ttl=255 time=12.060 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.10.20.1 icmp_seq=4 ttl=255 time=15.605 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.10.20.1 icmp_seq=5 ttl=255 time=6.890 ms (ICMP type:3, code:13, Communication administratively prohibited)

DCEE-PC-01> trace 10.10.40.10
trace to 10.10.40.10, 8 hops max, press Ctrl+C to stop
 1  *10.10.20.1  11.798 ms (ICMP type:3, code:13, Communication administratively prohibited) *
```

DCEE Users can access to the DIS WEB/DNS Servers



DMME users inside the same vlan

```
DMME-PC-01> ping 10.10.30.11

84 bytes from 10.10.30.11 icmp_seq=1 ttl=64 time=3.807 ms
84 bytes from 10.10.30.11 icmp_seq=2 ttl=64 time=5.566 ms
84 bytes from 10.10.30.11 icmp_seq=3 ttl=64 time=3.819 ms
84 bytes from 10.10.30.11 icmp_seq=4 ttl=64 time=20.166 ms
84 bytes from 10.10.30.11 icmp_seq=5 ttl=64 time=2.152 ms

DMME-PC-01> trace 10.10.30.11
trace to 10.10.30.11, 8 hops max, press Ctrl+C to stop
 1  *10.10.30.11  3.134 ms (ICMP type:3, code:3, Destination port unreachable)

DMME-PC-01> trace 10.10.30.10
traceroute to 10.10.30.10, 8 hops max
 1 10.10.30.10    0.001 ms
```

DMME to DEIE

```
DMME-PC-01> ping 10.10.10.10

*10.10.30.1 icmp_seq=1 ttl=255 time=10.011 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.10.30.1 icmp_seq=2 ttl=255 time=12.791 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.10.30.1 icmp_seq=3 ttl=255 time=6.779 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.10.30.1 icmp_seq=4 ttl=255 time=10.537 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.10.30.1 icmp_seq=5 ttl=255 time=5.886 ms (ICMP type:3, code:13, Communication administratively prohibited)

DMME-PC-01> trace 10.10.10.10
trace to 10.10.10.10, 8 hops max, press Ctrl+C to stop
 1  *10.10.30.1  5.841 ms (ICMP type:3, code:13, Communication administratively prohibited) *
```

DMME to DCEE

```
DMME-PC-01> ping 10.10.20.10

*10.10.30.1 icmp_seq=1 ttl=255 time=8.470 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.10.30.1 icmp_seq=2 ttl=255 time=6.581 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.10.30.1 icmp_seq=3 ttl=255 time=9.260 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.10.30.1 icmp_seq=4 ttl=255 time=5.874 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.10.30.1 icmp_seq=5 ttl=255 time=6.802 ms (ICMP type:3, code:13, Communication administratively prohibited)

DMME-PC-01> trace 10.10.20.10
trace to 10.10.20.10, 8 hops max, press Ctrl+C to stop
 1  *10.10.30.1 11.615 ms (ICMP type:3, code:13, Communication administratively prohibited) *
```

DMME to DIS

```
DMME-PC-01> ping 10.10.40.10

*10.10.30.1 icmp_seq=1 ttl=255 time=8.818 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.10.30.1 icmp_seq=2 ttl=255 time=4.952 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.10.30.1 icmp_seq=3 ttl=255 time=3.598 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.10.30.1 icmp_seq=4 ttl=255 time=7.386 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.10.30.1 icmp_seq=5 ttl=255 time=21.388 ms (ICMP type:3, code:13, Communication administratively prohibited)

DMME-PC-01> trace 10.10.40.10
trace to 10.10.40.10, 8 hops max, press Ctrl+C to stop
 1  *10.10.30.1 10.368 ms (ICMP type:3, code:13, Communication administratively prohibited) *
```

DIS users inside the same vlan

```
DIS-PC-01> ping 10.10.40.11

84 bytes from 10.10.40.11 icmp_seq=1 ttl=64 time=4.918 ms
84 bytes from 10.10.40.11 icmp_seq=2 ttl=64 time=8.505 ms
84 bytes from 10.10.40.11 icmp_seq=3 ttl=64 time=2.548 ms
84 bytes from 10.10.40.11 icmp_seq=4 ttl=64 time=2.625 ms
84 bytes from 10.10.40.11 icmp_seq=5 ttl=64 time=4.975 ms

DIS-PC-01> trace 10.10.40.11
trace to 10.10.40.11, 8 hops max, press Ctrl+C to stop
 1  *10.10.40.11  9.143 ms (ICMP type:3, code:3, Destination port unreachable)

DIS-PC-01> trace 10.10.40.10
traceroute to 10.10.40.10, 8 hops max
 1 10.10.40.10    0.001 ms
```

DIS to DEIE

```
DIS-PC-01> ping 10.10.10.10

84 bytes from 10.10.10.10 icmp_seq=1 ttl=62 time=23.614 ms
84 bytes from 10.10.10.10 icmp_seq=2 ttl=62 time=11.901 ms
84 bytes from 10.10.10.10 icmp_seq=3 ttl=62 time=33.627 ms
84 bytes from 10.10.10.10 icmp_seq=4 ttl=62 time=23.313 ms
84 bytes from 10.10.10.10 icmp_seq=5 ttl=62 time=14.980 ms

DIS-PC-01> trace 10.10.10.10
trace to 10.10.10.10, 8 hops max, press Ctrl+C to stop
 1  10.10.40.1  6.806 ms 7.185 ms 5.827 ms
 2  10.255.0.10 12.779 ms 15.477 ms 10.654 ms
 3  *10.10.10.10 16.872 ms (ICMP type:3, code:3, Destination port unreachable)
```

DIS to DCEE

```
DIS-PC-01> ping 10.10.20.10

10.10.20.10 icmp_seq=1 timeout
10.10.20.10 icmp_seq=2 timeout
10.10.20.10 icmp_seq=3 timeout
10.10.20.10 icmp_seq=4 timeout
10.10.20.10 icmp_seq=5 timeout

DIS-PC-01> trace 10.10.20.10
trace to 10.10.20.10, 8 hops max, press Ctrl+C to stop
 1  10.10.40.1  4.920 ms 7.161 ms 5.291 ms
 2  10.255.0.14 14.115 ms 10.727 ms 10.416 ms
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
```

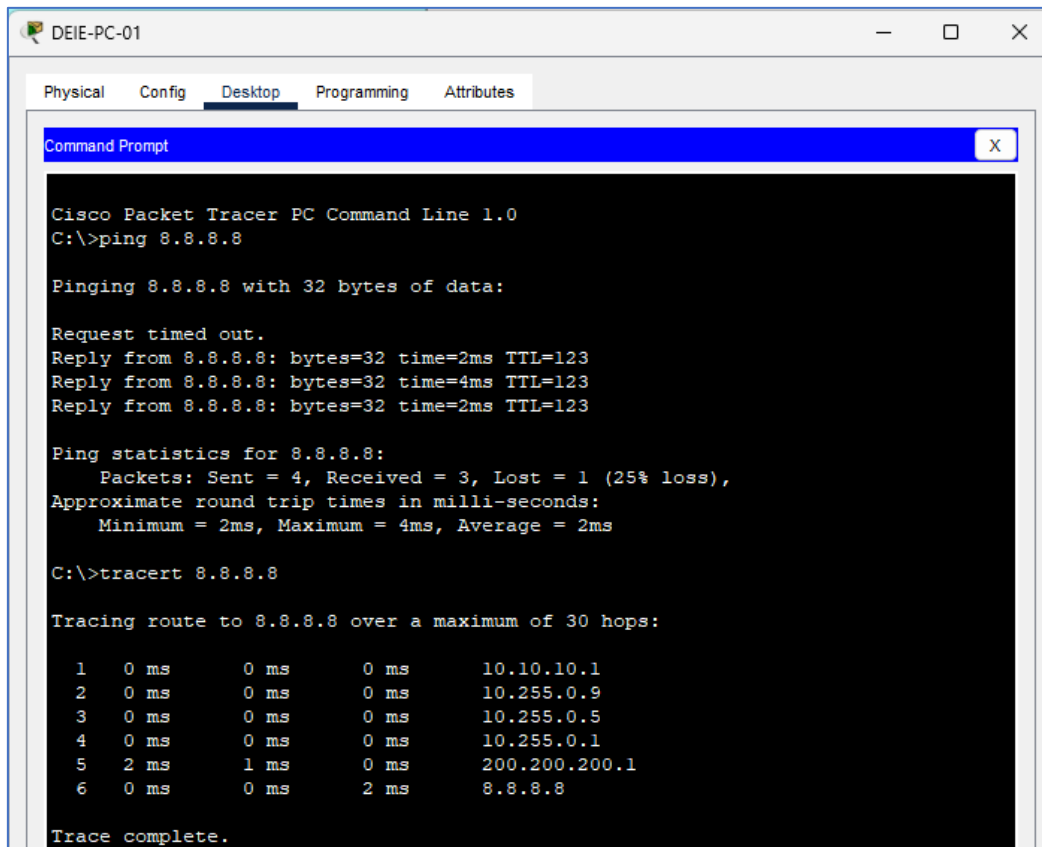
DIS to DMME

```
DIS-PC-01> ping 10.10.30.10

10.10.30.10 icmp_seq=1 timeout
10.10.30.10 icmp_seq=2 timeout
10.10.30.10 icmp_seq=3 timeout
10.10.30.10 icmp_seq=4 timeout
10.10.30.10 icmp_seq=5 timeout

DIS-PC-01> trace 10.10.30.10
trace to 10.10.30.10, 8 hops max, press Ctrl+C to stop
 1  10.10.40.1 13.631 ms 5.954 ms 7.554 ms
 2  10.255.0.18 18.991 ms 12.120 ms 9.919 ms
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
```

DEIE can access to the Internet through PAT



The screenshot shows a Cisco Packet Tracer PC Command Line window for DEIE-PC-01. The 'Desktop' tab is selected. The Command Prompt displays the following output:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Request timed out.
Reply from 8.8.8.8: bytes=32 time=2ms TTL=123
Reply from 8.8.8.8: bytes=32 time=4ms TTL=123
Reply from 8.8.8.8: bytes=32 time=2ms TTL=123

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 4ms, Average = 2ms

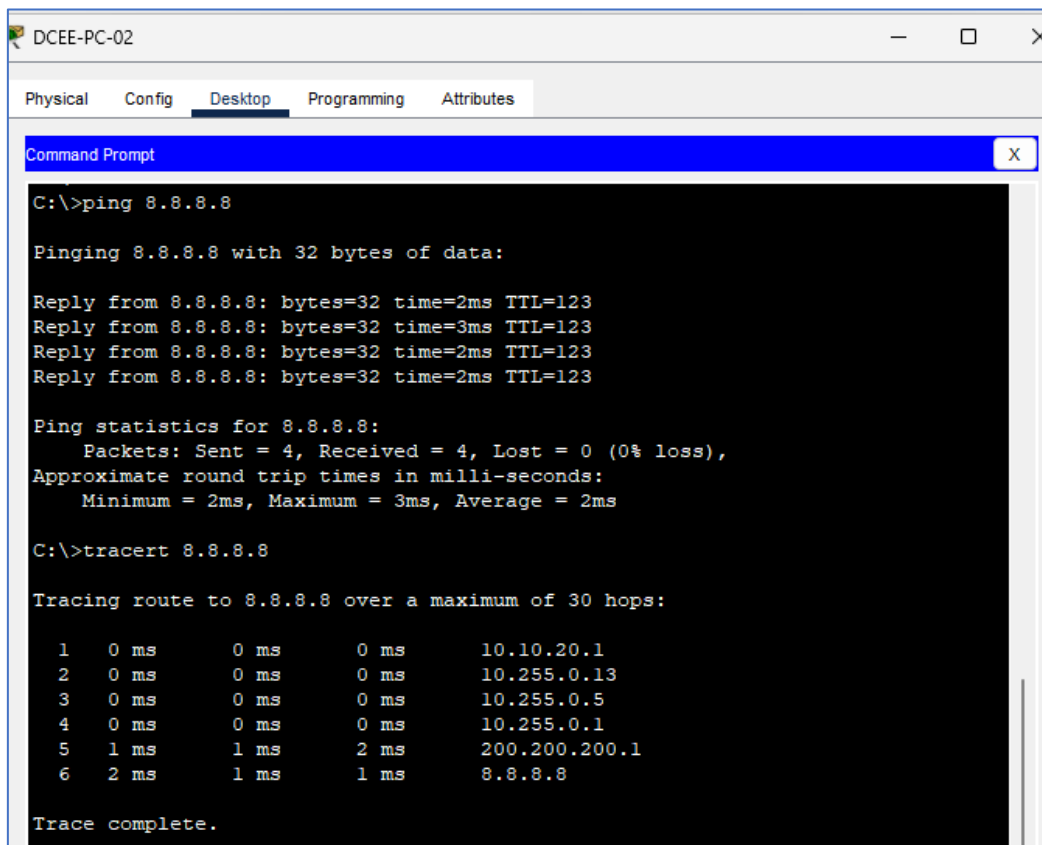
C:\>tracert 8.8.8.8

Tracing route to 8.8.8.8 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    10.10.10.1
  2  0 ms    0 ms    0 ms    10.255.0.9
  3  0 ms    0 ms    0 ms    10.255.0.5
  4  0 ms    0 ms    0 ms    10.255.0.1
  5  2 ms    1 ms    0 ms    200.200.200.1
  6  0 ms    0 ms    2 ms    8.8.8.8

Trace complete.
```

DCEE can access to the Internet through PAT



The screenshot shows a Cisco Packet Tracer PC Command Line window for DCEE-PC-02. The 'Desktop' tab is selected. The Command Prompt displays the following output:

```
C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time=2ms TTL=123
Reply from 8.8.8.8: bytes=32 time=3ms TTL=123
Reply from 8.8.8.8: bytes=32 time=2ms TTL=123
Reply from 8.8.8.8: bytes=32 time=2ms TTL=123

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

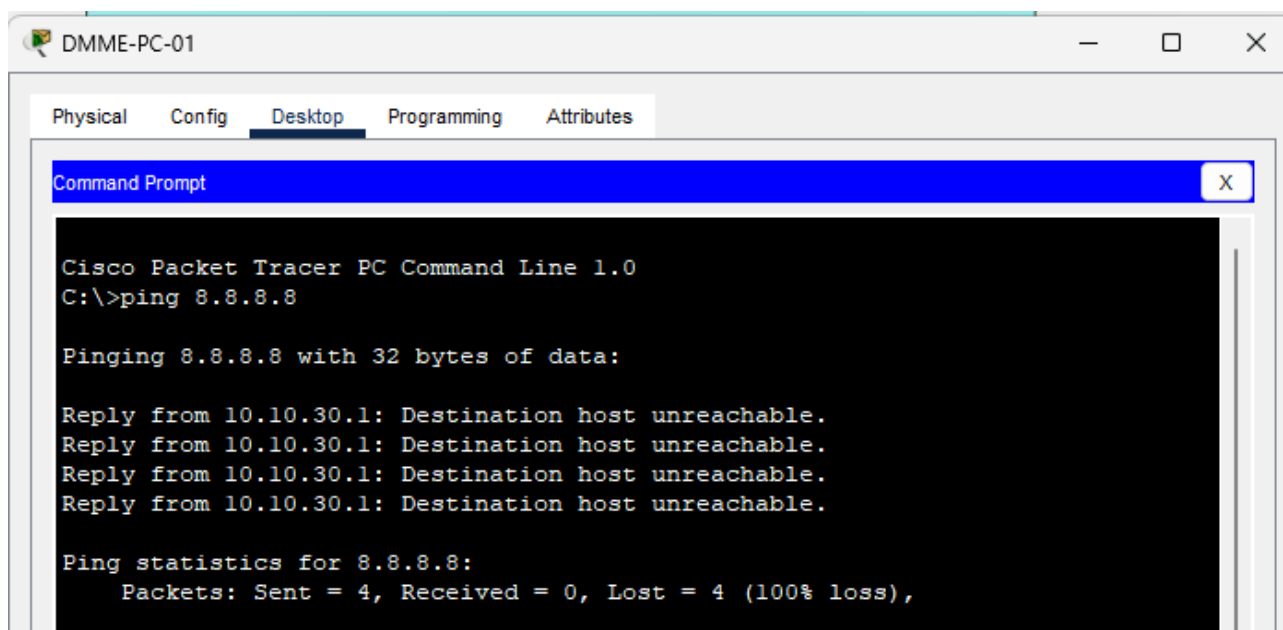
C:\>tracert 8.8.8.8

Tracing route to 8.8.8.8 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    10.10.20.1
  2  0 ms    0 ms    0 ms    10.255.0.13
  3  0 ms    0 ms    0 ms    10.255.0.5
  4  0 ms    0 ms    0 ms    10.255.0.1
  5  1 ms    1 ms    2 ms    200.200.200.1
  6  2 ms    1 ms    1 ms    8.8.8.8

Trace complete.
```


DMME can't access to the Internet through PAT (Since extended ACL)



DCEE can access to the Internet through PAT

